Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Core Pure 2 – Hyperbolic Functions

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Define $\sinh x$ and $\cosh x$ in terms of exponentials. **[2]**

2. Prove that $\cosh^2 x - \sinh^2 x = 1$. **[3]**

3. Find $\dfrac{d}{dx}(\sinh x)$ and $\dfrac{d}{dx}(\cosh x)$. **[3]**

4. Find $\int \sinh x \, dx$ and $\int \cosh x \, dx$. **[3]**

5. Solve $\sinh x = 0$. **[2]**

6. Solve $\cosh x = 1$. **[2]**

7. Express $\sinh 2x$ in terms of $\sinh x$ and $\cosh x$. **[3]**

8. Find $\tanh x$ in terms of $\sinh x$ and $\cosh x$. **[1]**

9. Prove that $1 - \tanh^2 x = \operatorname{sech}^2 x$. **[3]**

10. Find the inverse of $y = \sinh x$. **[3]**

11. Differentiate $\operatorname{arsinh} x$. **[3]**

12. Differentiate $\operatorname{arcosh} x$. **[3]**

13. Evaluate $\sinh(\ln 2)$. **[3]**

14. Evaluate $\cosh(\ln 3)$. **[3]**

15. Sketch $y = \sinh x$ and $y = \cosh x$. **[3]**

Section B – Hard (Q16–25)

16. Solve $\cosh x = 2$. **[4]**

17. Solve $\sinh x = 3$. **[3]**

18. Prove $\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$. **[4]**

19. Find $\int \operatorname{sech} x \, \mathrm{d}x$. **[4]**

20. Solve $3\cosh x + 4\sinh x = 5$. **[5]**

21. Find the Maclaurin series for $\sinh x$ up to the term in x^5. **[4]**

22. Show that $\cosh 2x = 2\cosh^2 x - 1$. **[3]**

23. Find the coordinates of the minimum point on $y = \cosh x$. **[3]**

24. Integrate $\int \frac{1}{\sqrt{x^2+1}} \, \mathrm{d}x$ using a hyperbolic substitution. **[5]**

25. Solve $\tanh x = \frac{1}{2}$. **[3]**

End of Worksheet • *Show all working clearly.*